

SMART MULTI PARAMETER INDUSTRIAL SAFETY SYSTEM

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Abstract:

The system is used to increase the safety of an industry by continuously monitoring multiple parameters. The industrial accidents cause risk to human life, losses and equipment damages. By using the various sensors and microcontroller it can detect dangerous conditions and gives alert through alarms. This system can provide or offers workplace safety, cost effective and more secure working environment.

Introduction:

Industrial safety plays a vital role in ensuring the protection of workers, equipment, and the environment. Modern industries are exposed to several risks, including gas leakage, fire, excessive temperature, smoke, and abnormal humidity levels, which can lead to severe accidents and economic losses. Therefore, continuous monitoring of these parameters is essential to maintain a safe and efficient working environment.

Smart multi parameter industrial safety system is designed to monitor the environment and operational system. Using a sensor it can detect abnormal conditions or issue like high temperature, gas leakage, humidity, etc.

The system can give immediate alert and alarm minimizing the risk of accidents.

Problem statement:

Industrial environments are prone to various risks such as gas leakage, fire outbreaks, excessive temperature, smoke generation, and abnormal humidity levels. Traditional safety systems often monitor only a single parameter and may not provide timely warnings during emergency situations. This limitation can result in equipment damage, production losses, environmental hazards, and threats to human life. Therefore, there is a need for an intelligent and reliable system capable of continuously monitoring multiple parameters simultaneously and providing real-time alerts to ensure a safe and secure industrial environment.

Approach:

To overcome the limitations smart multi parameter industrial system is approached. Whenever the parameter reaches above the safety threshold the system immediately sends the alarm to alert workers. This automated monitoring and alert mechanism enables quick response to unsafe conditions, thereby reducing the risk of accidents.

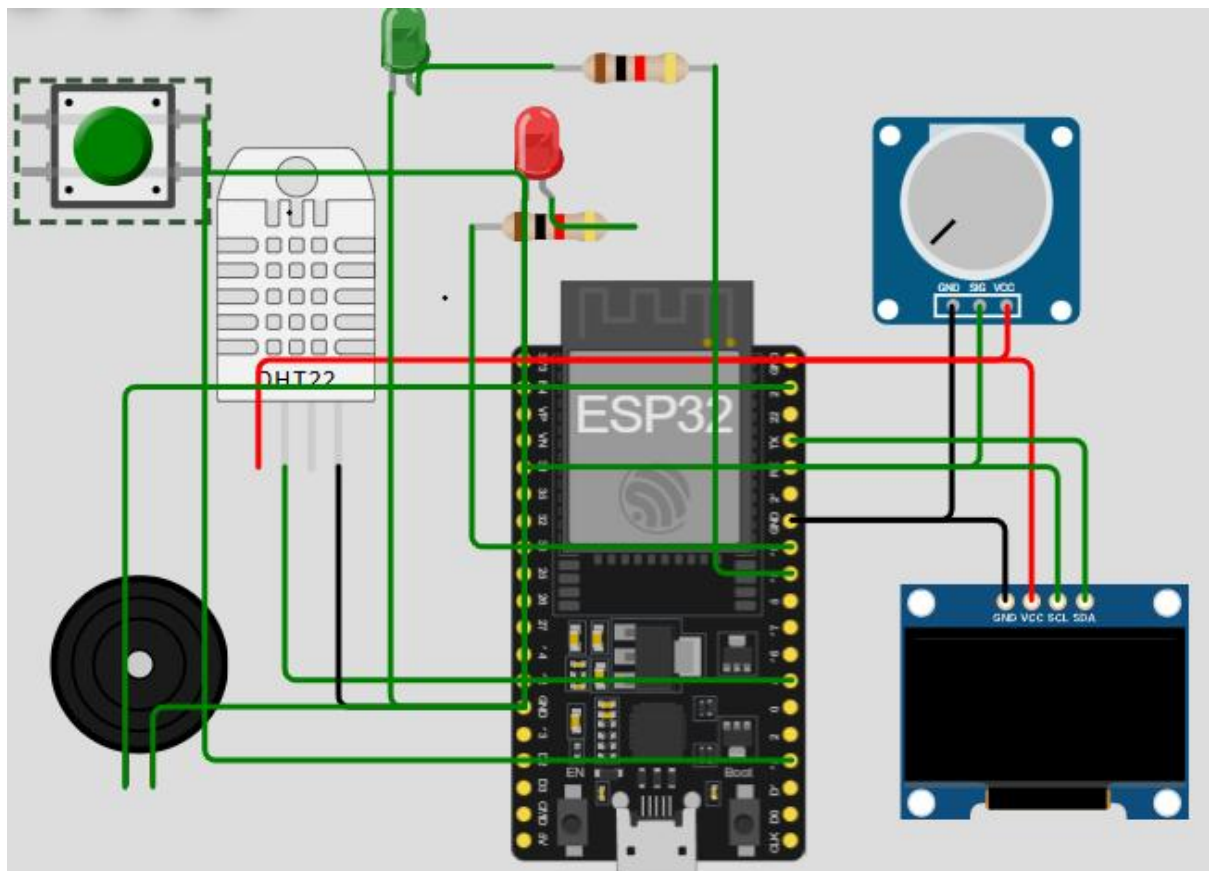
Objectives:

To design and develop an IoT-based Smart Multi-Parameter Industrial Safety System using ESP32 that continuously monitors temperature, humidity, gas leakage, and other environmental parameters, and provides real-time alerts to ensure workplace safety and prevent industrial accidents.

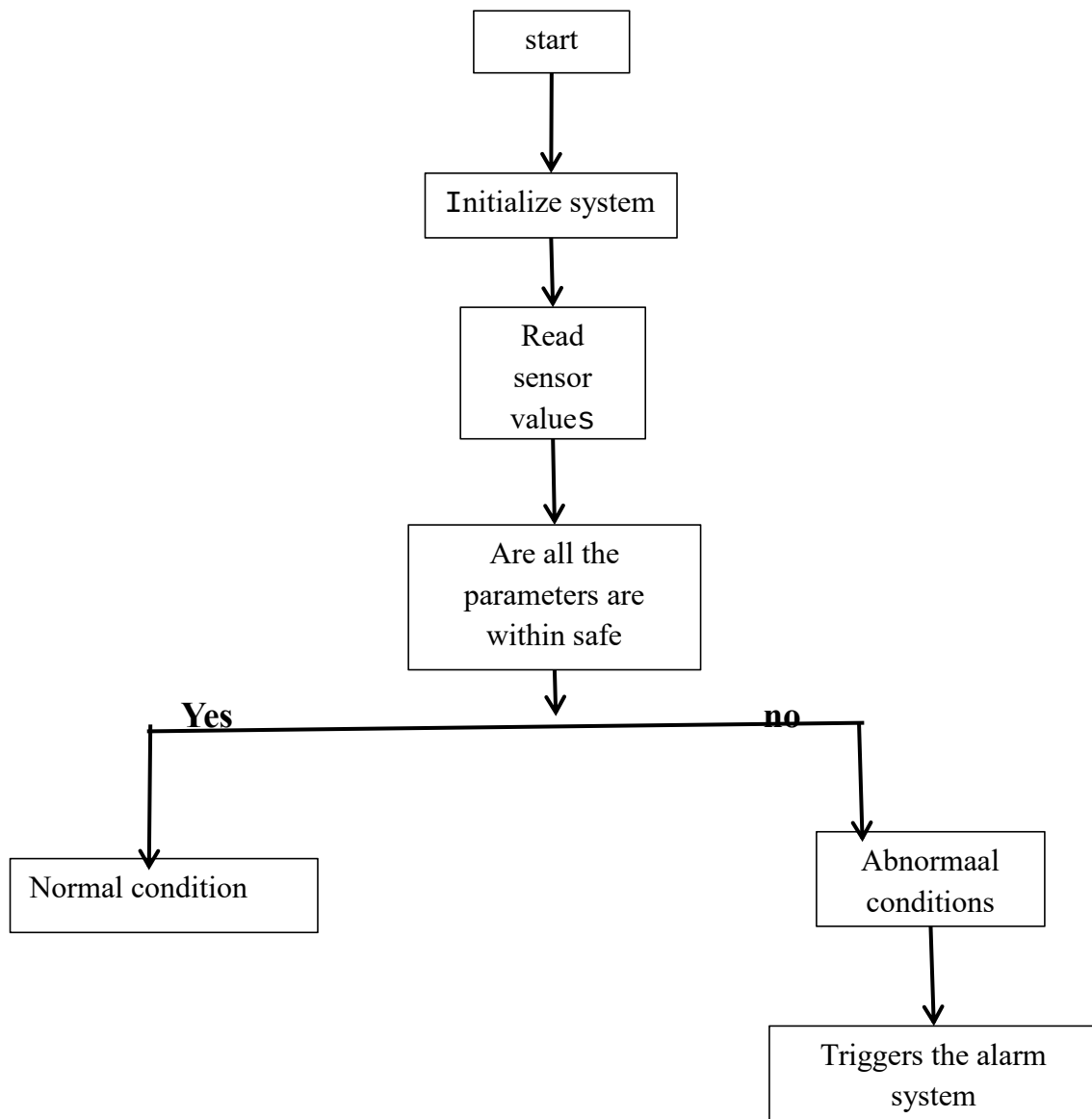
Components required:

- ESP32
- DHT22 Sensor
- MQ-2 Gas Sensor
- Red LED
- Green LED
- Buzzer
- Breadboard and Jumper Wires

Circuit diagram:



Flow chart:



System operation:

1. Sensors collect data.
2. ESP32 accepts and processes the data.
3. Sensor values are compared with defined safety limits.
4. If values are within safe limits:
 - Green LED remains ON.
 - System indicates normal operation.
5. If abnormal conditions are detected:
 - Red LED turns ON.
 - Buzzer activates.
 - Warning message is displayed or transmitted.

Working:

The Smart Multi-Parameter Industrial Safety System is an IoT-based safety monitoring system developed using an ESP32 microcontroller. The system continuously monitors important industrial safety parameters such as temperature, humidity, and gas concentration using sensors connected to the ESP32.

The **DHT22 sensor** measures temperature and humidity levels, while the MQ-2 gas sensor detects the presence of hazardous gases, smoke, or gas leakage in the industrial environment. The sensor data is continuously collected and processed by the ESP32.

The measured values are compared with predefined safety threshold levels. Under normal conditions, the system indicates safe operation through a green LED. When any parameter exceeds the safe limit, such as high temperature or gas leakage, the ESP32 immediately activates a red LED and buzzer to alert workers and supervisors about the potential hazard.

Challenges:

The primary challenge is ensuring accurate and reliable real-time monitoring

- Sensors may produce inaccurate readings due to environmental changes
- Temporary fluctuations in temperature, humidity, or gas concentration can cause unnecessary alerts.

Advantages:

Enhances Workplace Safety

- Continuously monitors industrial conditions and helps protect workers

Early Hazard Detection

- Detects gas leaks, excessive temperature, and abnormal environmental conditions

Real-Time Monitoring

- Provides continuous monitoring

Immediate Alert System

- Activates LEDs and buzzers instantly when unsafe conditions are detected.

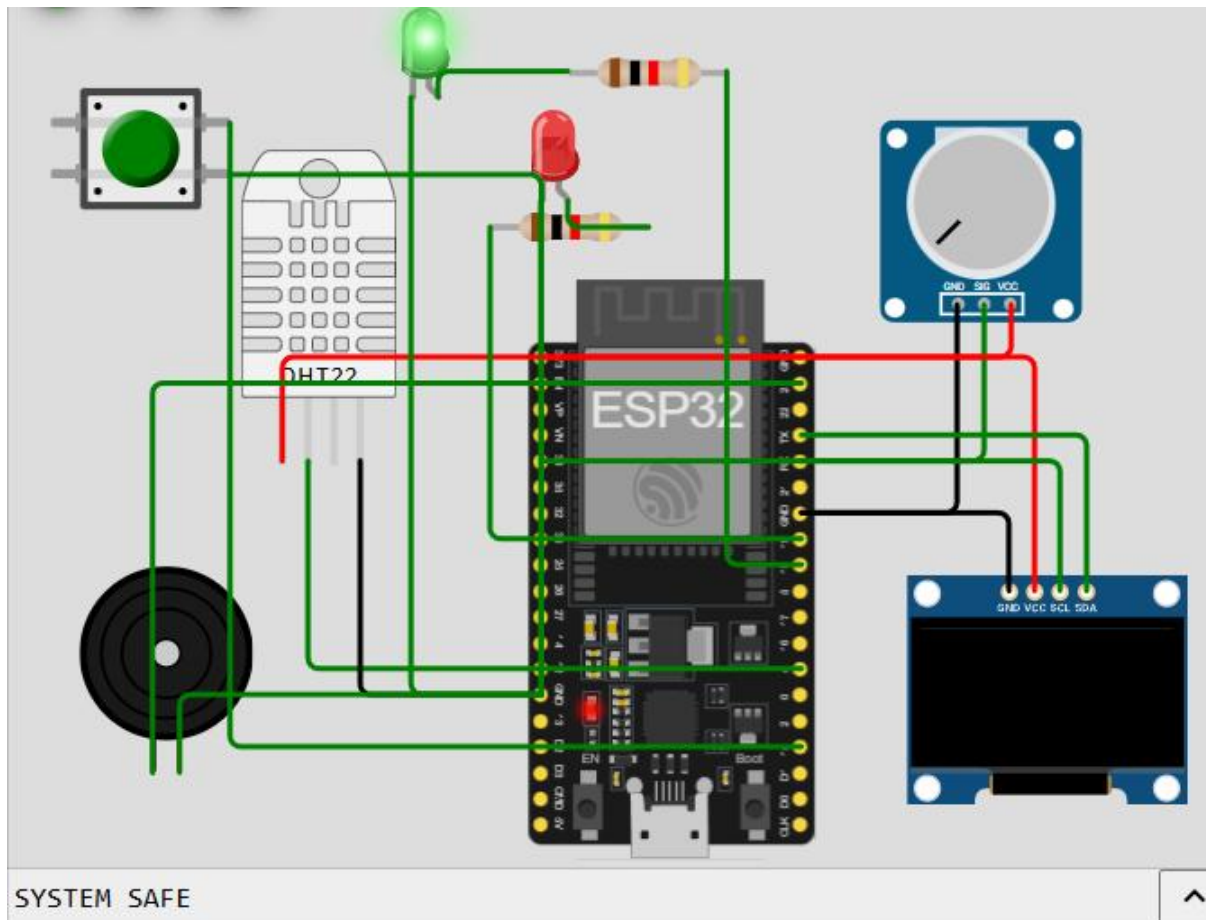
Reduces Industrial Accidents

- Helps prevent accidents, injuries, and equipment damage

Future scope:

The Smart Multi-Parameter Industrial Safety System can be further enhanced by integrating cloud computing, artificial intelligence, mobile applications, and advanced sensing technologies to provide intelligent, automated, and predictive industrial safety management.

Output:



Conclusions:

This project demonstrates how modern IoT technology can be utilized to create a low-cost, efficient, and automated industrial safety system. It contributes to reducing risks, protecting workers, minimizing equipment damage, and improving overall industrial safety management. With further enhancements such as cloud connectivity and remote monitoring, the system can be expanded into a more comprehensive industrial safety solution for future applications.

Reference:

Wokwi, *Wokwi Online Electronics Simulator Documentation*

GitHub Documentation